DS-604 Project Documentation

Formula 1 World Championship (1950–2020): Database Design and Performance Analysis

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1. Dataset Identification

Dataset Name: Formula 1 World Championship 1950–2020
Source: Kaggle
Dataset Link: https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020

The dataset contains historical data related to the Formula 1 World Championship from 1950 to 2020. It includes information about races, drivers, constructors (teams), circuits, race results, qualifying sessions, lap times, pit stops, and championship standings.

We selected this dataset because the data is divided into multiple files that are related to each other. This makes it suitable for a DBMS project, where the data can be organized into different tables and connected using primary and foreign keys.

2. Dataset Description

The dataset contains information collected from Formula 1 races over several decades. Each file represents a different type of information related to Formula 1.

File	Description
Drivers	Contains details about Formula 1 drivers such as driver ID, name, date of birth, nationality, and other driver information.
Constructors	Contains information about Formula 1 teams, including their names and nationalities.
Circuits	Contains information about racing circuits, including their names, locations, countries, and geographical coordinates.
Races	Contains details about individual races, including the season, race round, date, race name, and circuit.
Results	Contains the results of drivers in individual races, including their grid position, finishing position, points, laps, and fastest-lap information.
Qualifying	Contains the qualifying results of drivers before races.
Lap Times	Contains information about the time taken by drivers on individual laps.
Pit Stops	Contains information about pit stops made during races, including the lap and duration of the stop.
Driver Standings	Contains information about drivers' championship standings after races.
Constructor Standings	Contains information about constructors' championship standings.
Constructor Results	Contains the results and points obtained by constructors in races.
Seasons	Contains information about the different Formula 1 seasons.

Status	Contains different statuses describing how a driver finished or left a race.
Sprint Results	Contains information about sprint race results.

3. Relationships Between the Data

The different files in the dataset are connected through common ID fields. These IDs will be used as primary keys and foreign keys when designing the relational database.

1. **Drivers → Results —** A driver can participate in many races, so one driver can have multiple records in the Results table. The driverId connects the two tables.
2. **Constructors → Results —** A constructor (team) can have multiple race results. The constructorId connects the Constructors and Results tables.
3. **Races → Results —** One race can have results for many drivers. The raceId connects the Races and Results tables.
4. **Circuits → Races —** A circuit can host multiple races over different years. The circuitId connects the Circuits and Races tables.
5. **Drivers → Qualifying —** A driver can participate in qualifying sessions for many races. The driverId connects the Drivers and Qualifying tables.
6. **Races → Qualifying —** Each qualifying record belongs to a particular race. The raceId connects the Races and Qualifying tables.
7. **Drivers → Lap Times —** A driver can have multiple lap-time records across different races. The driverId connects the Drivers and Lap Times tables.
8. **Races → Lap Times —** A race contains multiple lap-time records for participating drivers. The raceId connects the Races and Lap Times tables.
9. **Drivers → Pit Stops —** A driver can make multiple pit stops during races. The driverId connects the Drivers and Pit Stops tables.
10. **Races → Pit Stops —** A race can contain multiple pit stops made by different drivers. The raceId connects the Races and Pit Stops tables.
11. **Drivers → Driver Standings —** A driver can have multiple championship-standing records throughout a season. The driverId connects the Drivers and Driver Standings tables.
12. **Races → Driver Standings —** Driver standings are recorded after particular races. The raceId connects the Races and Driver Standings tables.
13. **Constructors → Constructor Standings —** A constructor can have multiple championship-standing records. The constructorId connects the Constructors and Constructor Standings tables.
14. **Races → Constructor Standings —** Constructor standings are associated with particular races. The raceId connects the Races and Constructor Standings tables.
15. **Constructors → Constructor Results —** A constructor can have results for many races. The constructorId connects the Constructors and Constructor Results tables.

16. **Races → Constructor Results —** Constructor results belong to particular races. The raceId connects the Races and Constructor Results tables.
17. **Status → Results —** Each race result has a status, such as finished, retired, or disqualified. The statusId connects the Status and Results tables.
18. **Drivers → Sprint Results —** A driver can participate in multiple sprint races. The driverId connects the Drivers and Sprint Results tables.
19. **Races → Sprint Results —** Each sprint result belongs to a particular race. The raceId connects the Races and Sprint Results tables.
20. **Constructors → Sprint Results —** A sprint result is also associated with the constructor/team of the driver. The constructorId connects the Constructors and Sprint Results tables.
21. **Seasons → Races —** Each race belongs to a particular Formula 1 season. The year field connects the season information with the corresponding races.

Overall Relationship Structure

Drivers → Results ← Constructors → Races ← Circuits → (Qualifying, Lap Times, Pit Stops, Driver Standings, Constructor Standings, Constructor Results, Sprint Results)

The Status table is connected to Results through statusId, while the Seasons information is associated with races through the season/year.

These relationships make the dataset suitable for designing a relational database because the information can be divided into separate tables while maintaining connections between them using primary keys and foreign keys.

4. Analytical Objectives

The broad objectives guiding our query design are to:

1. Identify the most successful drivers and constructors across F1 history using win, points, and championship metrics.
2. Analyse performance trends over time, including lap-time evolution and constructor dominance across decades.
3. Study the relationship between qualifying/grid position and race outcome.
4. Quantify operational aspects of racing such as pit-stop efficiency and race-retirement (DNF) patterns.
5. Explore demographic and geographic dimensions, such as nationality of winners and home-circuit advantage.
6. Compare performance across related but distinct formats, such as sprint races versus main Grands Prix.

Each objective above is deliberately mapped to two to three of the fifteen queries in Section 5, ensuring balanced coverage across all 14 tables rather than over-concentrating on any single one (e.g., results).

5. Tentative Analytical Queries

Fifteen tentative analytical queries have been identified, each mapped to the source tables required to answer it and to the key SQL construct(s) it is expected to exercise. This demonstrates coverage of core relational operations — multi-table joins, grouping and aggregation, filtering, and window functions.

	Query Theme	Analytical Question	Tables Involved	Key SQL / Technique
1	Career wins	Which drivers have the highest number of career race wins, and how does the top 10 list look across all seasons (1950–2020)?	drivers, results	GROUP BY, COUNT, ORDER BY / LIMIT
2	Constructor dominance	Which constructors have won the most Constructors' Championships, and in which eras did they dominate?	constructor_standings, races, constructors	JOIN, subquery on season-end standings
3	All-time points leaders	Who are the top 10 drivers by total career championship points?	drivers, driver_standings	JOIN, GROUP BY, aggregate SUM/MAX
4	Pit-stop efficiency	What is the average pit-stop duration per constructor per season, and which team is consistently the fastest in the pits?	pit_stops, results, races, constructors	Multi-way JOIN, AVG, GROUP BY
5	Busiest circuits	Which circuits have hosted the highest number of Grands Prix, and how has the race calendar evolved geographically?	circuits, races	JOIN, COUNT, GROUP BY
6	Lap-time evolution	How has the average lap time at a given circuit changed over the decades (car performance evolution)?	lap_times, races, circuits	JOIN, AVG, GROUP BY (decade bucket)
7	Qualifying vs. race outcome	What is the correlation between a driver's qualifying (grid) position and their final race position?	qualifying, results, races	JOIN on (raceId, driverId), derived column
8	Winners by nationality	What is the nationality distribution of race winners, and which countries have produced the most successful drivers?	drivers, results	JOIN, COUNT, GROUP BY
9	DNF / retirement analysis	Which constructors have the highest number of race retirements (DNFs), and what are the most common causes (engine, accident, gearbox, etc.)?	status, results, constructors	JOIN, GROUP BY, CASE / filtering
10	Season points progression	How do the championship points of the top 5 drivers in a given season progress race-by-race?	driver_standings, races, drivers	JOIN, window function (running total/rank)

11	Fastest-lap leaders	Which drivers and constructors hold the most 'fastest lap of the race' awards, and how are these distributed across seasons?	results, drivers, constructors, races	Filter on rank = 1, JOIN, GROUP BY
12	Home-race advantage	Do drivers statistically perform better (higher finishing position) at their home country Grand Prix compared to their season average?	drivers, races, circuits, results	Self-referencing comparison, CASE, AVG
13	Decade-wise team dominance	What share of total points did each constructor earn per decade, highlighting the rise and fall of dominant teams?	constructor_standings, races, constructors	JOIN, GROUP BY, window function (% share)
14	Grid position vs. points scored	Does starting grid position show a statistically meaningful relationship with points scored in the race?	results, races	JOIN, AVG, GROUP BY (grid buckets)
15	Sprint vs. Grand Prix performance	For seasons with sprint races (2021 onward), how does a driver's sprint result compare with their main-race result on the same weekend?	sprint_results, results, races, drivers	Multi-table JOIN, comparative subquery